

7. Glucose is a stable compound and does not undergo dehydration easily. It indicates that not more than one hydroxyl group is bonded to a single carbon atom. Thus the five the hydroxyl groups are attached to five different carbon atoms and the sixth carbon is an aldehyde group.
8. The exact spacial arrangement of -OH groups was given by Emil Fischer as shown in Figure 14.4. The glucose is referred to as D(+) glucose as it has D configuration and is dextrorotatory.

Cyclic structure of glucose

Fischer identified that the open chain penta hydroxyl aldehyde structure of glucose, that he proposed, did not completely explain its chemical behaviour. Unlike simple aldehydes, glucose did not form crystalline bisulphite compound with sodium bisulphite. Glucose does not give Schiff's test and the penta acetate derivative of glucose was not oxidized by Tollen's reagent or Fehling's solution. This behaviour could not be explained by the open chain structure.

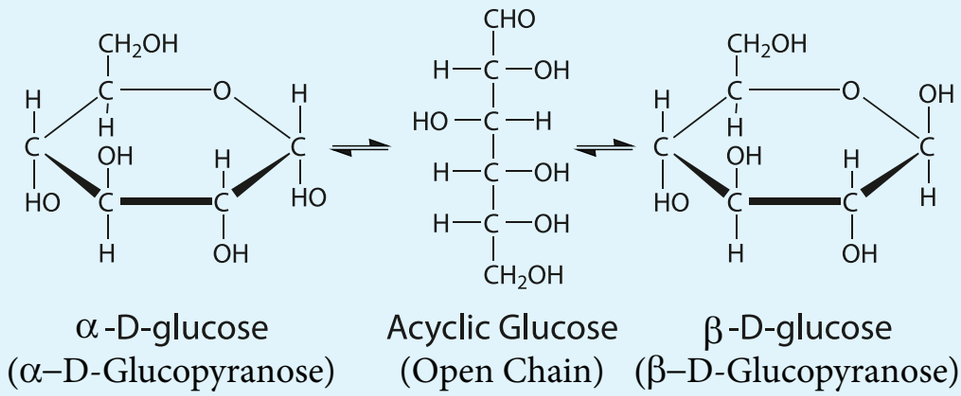


Figure 14.5 Cyclic structures of glucose

In addition, glucose is found to crystallise in two different forms depending upon the crystallisation conditions with different melting points (419 and 423 K). In order to explain these it was proposed that one of the hydroxyl group reacts with the aldehyde group to form a cyclic structure (hemiacetal form) as shown in figure 14.5. This also results in the conversion of the achiral aldehyde carbon into a chiral one leading to the possibility of two isomers. These two isomers differ only in the configuration of C1 carbon. These isomers are called **anomers.** The two anomeric forms of glucose are called α and β-forms. This cyclic structure of glucose is similar to pyran, a cyclic compound with 5 carbon and one oxygen atom, and hence is called pyranose form. The specific rotation of pure α- and β-(D) glucose are 112° & 18.7° respectively. However, when a pure form of any one of these sugars is dissolved in

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